

Section-A (25 Marks)

1. Point out different categories of visual aids for navigation at an airport. Explain taxiway marking with sketch. [2+3=5]

Answer:

Visual aids are provided to ensure safe and efficient aircraft movement. They are universally standardized to prevent runway invasion, eliminate taxiway confusion, and help pilots navigate final approaches and ground operations in all visibility conditions. Airports utilize broad categories of visual aids which are listed below:

i. Lighting systems:

These are electric lights used at night or in low visibility to define runways, taxiways, and approach paths such as runway edge and centerline lights, approach lighting systems (ALS), Precision Approach Path Indicator (PAPI), Visual Approach Slope Indicator (VASI), Visual Glideslope Indicators (VGSI) etc.

ii. Pavement markings:

Painted markings on the surface guide movement and positioning on runways, taxiways, and aprons such as runway centerlines and threshold markings, taxiway centerlines, touchdown zone markings etc. These markings are universally color-coded **white** for runway operations (landing and taking off) and **yellow** for taxiing and ground movements.

iii. Signals and boards:

Fixed signs provide information about location, direction, restrictions, and distances such as location signs, direction and destination signs, information signs etc.

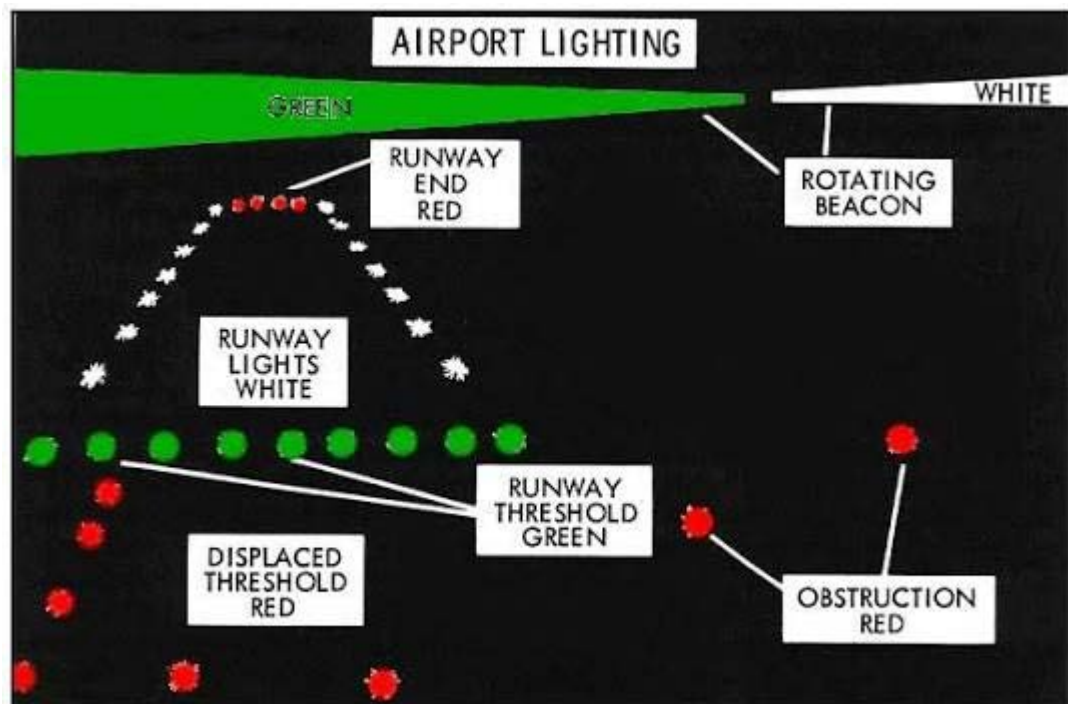


Figure 14-3 Airport Lighting

Taxiway markings:

Taxiway markings are yellow painted lines and symbols on the pavement that guide aircraft along taxiways and show where to stop, turn, or avoid entering. All taxiway markings are yellow which is easily distinguishable to pilots from runway markings.

i. Taxiway centerline marking:

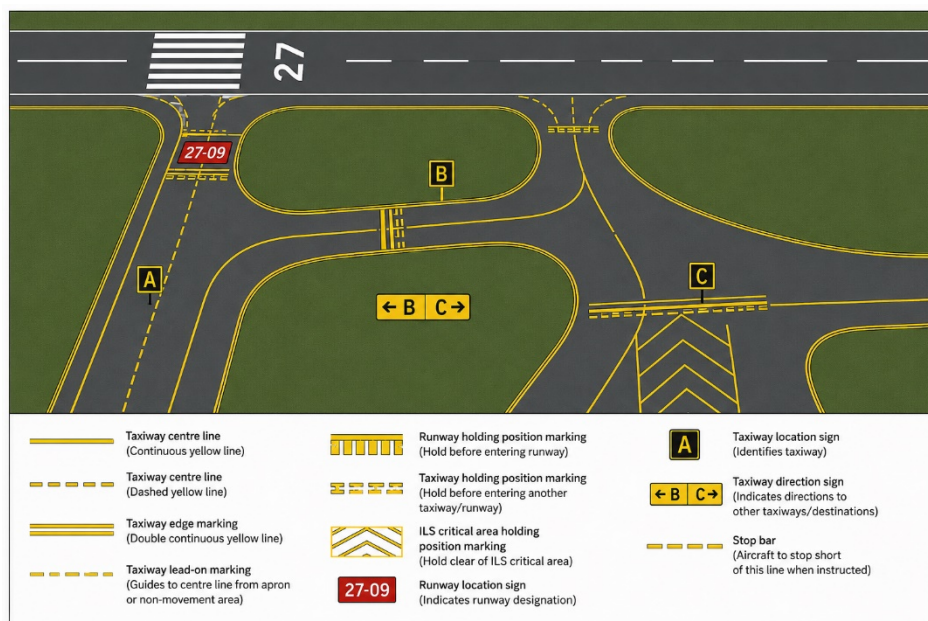
- A single continuous yellow line painted along the center of the taxiway.
- It provides guidance for the aircraft to remain in the center of the taxiway.
- At taxiway intersections, the line curves smoothly to indicate the correct turning path.

ii. Taxiway edge markings:

- These define the edge of the taxiway pavement.
- Continuous Double Yellow Line: Indicates the pavement beyond the line is not intended for aircraft use.
- Dashed Double Yellow Line: Indicates the pavement beyond the line may be used, such as a widened shoulder or adjoining paved area.

iii. Taxiway shoulder markings:

- Yellow stripes placed on taxiway shoulders.
- Indicates that the shoulder is not designed to support aircraft loads and should not be used for normal taxiing.



2. Describe briefly about International Civil Aviation Organization (ICAO)'s requirement for aerodrome design and operations. [5]

Answer:

ICAO's requirements for aerodrome design and operations are laid down mainly in Annex 14 "Aerodromes", which sets global minimum standards for how airports are planned, built, equipped, and operated to ensure safety and uniformity worldwide.

i. Scope and structure:

ICAO publishes Standards and Recommended Practices (SARPs) for aerodromes in Annex 14, Volume I: Aerodrome Design and Operations.

These SARPs cover physical characteristics, obstacle control, visual aids, rescue and firefighting, and operational procedures, and are complemented by detailed guidance in the Aerodrome Design Manual and PANS-Aerodromes (Doc 9981).

ii. Physical design of aerodomes:

Annex 14 specifies required runway and taxiway dimensions, separations, and clearances, based on an aerodrome reference code that depends on aircraft size and approach speed.

It also defines runway strips, runway end safety areas (RESA), stop ways, taxiway shoulders, aprons, and pavement characteristics, ensuring adequate bearing strength and safety margins for design aircraft.

iii. Obstacle limitation and safety areas:

The Annex defines Obstacle Limitation Surfaces (OLS) around the aerodrome (approach, take-off climb, transitional, inner horizontal, etc.) to control buildings, masts, trees, and terrain that might endanger aircraft.

It includes requirements for marking and lighting obstacles, and for maintaining clear and graded areas such as runway strips and RESA to reduce damage risk if aircraft overrun or veer off.

iv. Visual aids, signs and lightning:

ICAO specifies standardized markings, signs, and lighting systems for runways, taxiways, and aprons to provide uniform visual guidance.

Requirements cover runway and taxiway markings, approach and runway lighting, PAPI/VASI, taxiway centerline and stop-bar lights, apron floodlighting, and aerodrome beacons, with detailed photometric and color standards.

v. Operational and safety requirements:

Annex 14 includes provisions for aerodrome operations, such as inspection and maintenance of movement areas, friction measurement, snow and contamination control, wildlife hazard management, and emergency planning.

It defines categories and minimum levels of Rescue and Fire Fighting (RFF) services, including response times, equipment, and extinguishing agent quantities according to the largest aircraft served.

vi. Certification and state responsibilities:

ICAO requires states to establish a system for aerodrome certification, ensuring that international aerodromes comply with Annex 14 SARPs or that any differences are documented and notified to ICAO.

States must oversee operators' safety management, keep aeronautical information publications (AIP) up to date, and implement continuous compliance and safety oversight programmes.

3. Explain different structural components of a flexible pavement. How does flexible pavement differ from rigid pavement? Explain. [10]

Answer:

The pavements which have very low structural strength and are flexible in their structural behavior under the load are called flexible pavements. The flexible pavements layers reflect the deformation of the lower layers onto the surface of the layer by grain-to-grain contact. The strength of the pavement depends on the quality and thickness of each layer.

Structural components of flexible pavement:

i. Subgrade:

The **subgrade** is the natural soil or prepped embankment on which the pavement rests.

It is the foundation of the pavement which supports all the upper layers.

Subgrade of the pavement should have adequate bearing capacity, proper drainage, and good compaction.

Weak subgrades require stabilization or improvement before pavement construction.

Functions:

- Supports the entire pavement structure.
- Distributes loads to the ground.
- Provides a stable working platform.

ii. Sub-base course:

The **sub-base** is a layer of granular material placed above the subgrade and below the base course.

It is provided when the subgrade is weak or when additional structural support is needed.

Materials that form sub-base course:

- Granular soil
- Crushed stone
- Gravel
- Stabilized soil

Functions:

- Distributes wheel loads.
- Improves drainage.
- Prevents pumping of fine soil.
- Protects the subgrade from frost and moisture.

iii. Base course:

The **base course** is the principal structural layer located above the sub-base.

It is made of high-quality crushed aggregates or stabilized materials.

Materials:

- Crushed stone
- Water Bound Macadam (WBM)
- Wet Mix Macadam (WMM)
- Cement-treated base

Functions:

- Carries and distributes traffic loads.
- Provides structural strength.
- Reduces stress transmitted to the subgrade.
- Supports the surface course.

iv. Binder course/ Intermediate course:

The **binder course** lies between the base course and the wearing course.

It consists of bituminous material mixed with coarse aggregates.

Functions:

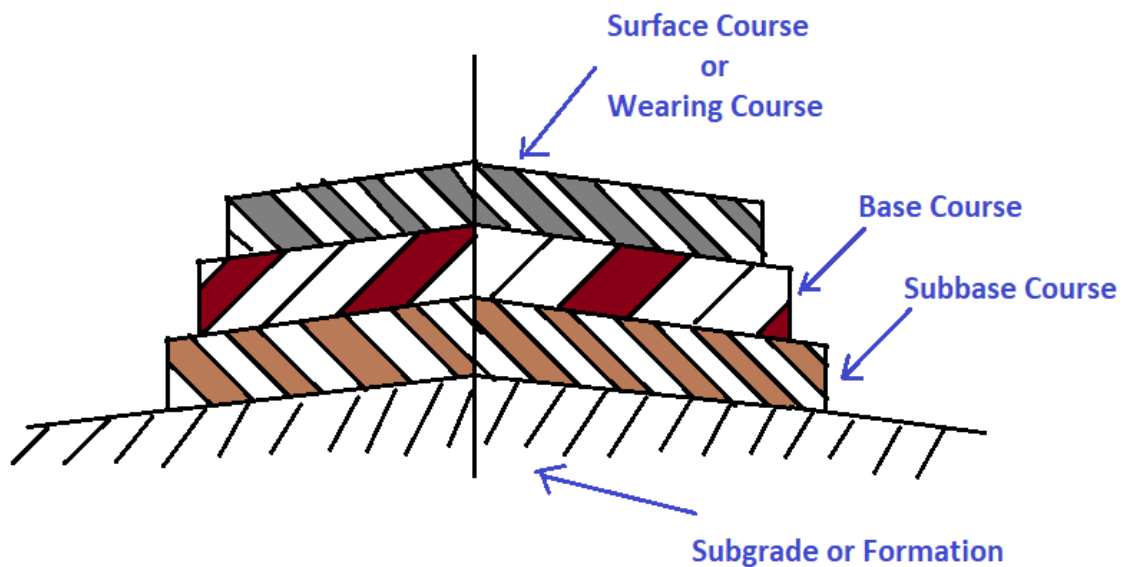
- Provides additional structural strength.
- Binds the surface and base layers.
- Distributes traffic stresses.
- Forms a stable platform for the wearing course.

v. **Surface/ wearing course:**

The **surface course** is the topmost layer directly in contact with traffic. It is made of dense bituminous concrete or asphalt concrete.

Functions:

- Provides a smooth riding surface.
- Resists wear and abrasion.
- Protects lower layers from water infiltration.
- Provides skid resistance and riding comfort.



Flexible pavement differs from rigid pavements, described in the table below:

Flexible pavements	Rigid pavements
i. Flexible pavements on the heavy load yield to excessive stresses resulting in local depression of the surface.	1. Rigid pavement under a heavy load ruptures, producing a crack to the surface.
2. Flexible pavement with subgrade of varying thickness will adjust itself to irregularities due to differential settlements.	ii. Rigid pavement with the subgrade of varying strength will not adjust to irregularities.
iii. Temperature variations due to atmospheric conditions do not produce stress.	3. Temperature variations produce heavy temperature stresses.
iv. Flexible pavement has self-healing properties.	4. Rigid pavement do not have self-healing properties.
v. Strength of flexible layer is by distributing the load over subgrade.	5. Strength of rigid layer is by bending action.

4. What is design consideration for airport terminal area? Explain. [10]

Answer:

Airport terminal area is the part of an airport where passengers, baggage, cargo, and aircraft interface. The terminal should be designed to provide safe, efficient, comfortable, and economical movement of passengers and aircraft while allowing for future expansion.

The design consideration for airport terminal area is explained below:

i. Passenger traffic forecast:

The terminal should be designed based on the peak hour passenger demand and projected future growth.

Passenger traffic forecast consideration is necessary as it:

- Prevents overcrowding.
- Allows future expansion.

ii. Terminal building layout:

The layout should ensure smooth passenger flow from entrance to boarding and from arrival to exit.

Terminal building layout should be considered in airport terminal area because it:

- Reduces delays and confusion.
- Improves passenger convenience.

iii. Aircraft Apron and gates:

The apron should provide adequate parking positions for aircraft.

Aircraft Apron and gates should be considered in airport terminal area because it:

- Ensures safe aircraft operations.
- Minimizes turnaround time.

iv. Baggage handling system:

Efficient baggage handling facilities should be provided.

Baggage handling system should be considered in airport terminal area because it:

- Reduces baggage delays.
- Improves operational efficiency.

v. Vehicular access and planning:

Easy access roads should connect the terminal to the city.

Vehicular access and planning should be considered in airport terminal area because it:

- Reduces traffic congestion.
- Improves accessibility.

vi. Safety and security:

Installation of CCTV surveillance.

Fire detection and firefighting systems.

Safety and security should be considered in airport terminal area as it:

- Protects passengers, staff, and airport infrastructure.

vii. Passenger facilities:

The terminal should include:

- Check-in counters
- Security screening
- Immigration and customs (international airports)
- Waiting lounges
- Restaurants and shops

Safety and security should be considered in airport terminal area as it:

- Enhances passenger comfort and convenience.

viii. Accessibility:

The terminal should be accessible to all users by providing:

- Ramps
- Elevators
- Escalators
- Wheelchair-accessible facilities

Accessibility should be considered in airport terminal area as it:

- Ensures universal accessibility.

ix. Environmental considerations:

Proper drainage and stormwater management.

Noise control measures.

Green building practices.

Environmental considerations should be considered in airport terminal area as it:

- Promotes sustainable airport operations.

x. Future expansion:

Structural design should allow additional gates, parking stands, and passenger facilities as demand increases.

Future expansion should be considered in airport terminal area as it:

- Reduces future reconstruction costs.
- Supports long-term airport development.

5. What is airport Master Planning? Elaborate the types of activities involved in the master plan process. [3+7=10]

Answer:

Airport Master Planning is a comprehensive and long-term planning process that guides the orderly development of an airport to meet present and future aviation demand. It provides a framework for the efficient use of land, facilities, and financial resources while ensuring safety, operational efficiency, environmental sustainability, and economic feasibility.

The master plan typically covers a planning period of **20–30 years** and serves as the basis for airport development, expansion, and investment decisions.

Objectives of Airport Master Planning:

- Forecast future air traffic demand.
- Plan for future expansion.
- Minimize environmental impacts.
- Estimate development costs and prioritize projects.
- Coordinate airport development with regional and national transportation plans.

Activities involved in the Airport Master Plan Process:

a. Inventory of existing facilities:

Data on the existing airport infrastructure must be collected, including:

- a. Runways and taxiways
- b. Aprons
- c. Terminal buildings
- d. Navigational aids
- e. Utilities
- f. Access roads
- g. Land use

The condition and capacity of current facilities must be properly assessed.

b. Aviation Demand Forecast:

The future passenger traffic, aircraft movements, and cargo demand is forecasted.

Short-term, medium-term, and long-term growth is evaluated.

Different factors such as population growth, tourism, economic development, and airline operations is properly considered.

c. Capacity Analysis:

It must be properly evaluated whether existing airport facilities can accommodate the forecasted demand.

The deficiencies is identified in:

- Runway capacity
- Taxiway system
- Terminal capacity
- Apron parking
- Car parking
- Air traffic control facilities

d. Development of airport alternatives:

Prepare different development options for airport expansion.

Compare alternatives based on:

- Safety
- Operational efficiency
- Cost
- Environmental impact
- Future expansion potential

e. Environmental assessment:

The environmental impacts of proposed developments are studied, including:

- Aircraft noise
- Air pollution
- Water pollution
- Land use changes
- Wildlife hazards

Then, the suitable mitigation measures are recommended.

f. Airport layout plan:

A detailed layout is prepared, showing:

- a. Runways
- b. Taxiways
- c. Aprons
- d. Terminal buildings
- e. Hangars
- f. Future expansion zones

The Airport Layout Plan serves as the primary drawing for future airport development.

g. Financial and economic analysis:

The construction, operation, and maintenance costs are estimated.
Identify possible funding sources are identified.
Cost-benefit and economic feasibility analyses are conducted.
The development projects based on available resources are prioritized.

h. Implementation plan:

The development program is divided into phases:

- Short-term (0–5 years)
- Medium-term (5–10 years)
- Long-term (10–20 or more years)

The project priorities and implementation schedules are established.

i. Monitoring and evaluation:

The master plan is reviewed regularly.
The forecasts and development priorities as traffic demand and airport requirements change are regularly updated.

It is to be ensured that the plan remains relevant throughout the planning period.

Hence, by forecasting future demand, evaluating existing facilities, and planning phased implementation, a master plan provides a clear roadmap for airport growth over the long term while supporting economic development and maintaining high standards of aviation safety.

6. According to the Civil Aviation Authority Act, 2053. Write functions, duties and power of Civil Aviation Authority of Nepal. [10]

Answer:

According to the Civil Aviation Authority Act, 2053 (1996), the Civil Aviation Authority of Nepal (CAAN) is established as an autonomous body to regulate, supervise and develop civil aviation in Nepal. Based on that Act, its functions, duties and powers can be grouped as follows:

i. Regulatory and safety functions:

- To regulate and supervise all civil aviation activities within Nepalese territory in accordance with national law and ICAO standards.
- To issue, renew, suspend or cancel licenses, ratings and certificates of aviation personnel (pilots, engineers, ATC, etc.) and airworthiness certificates for aircraft.
- To issue technical regulations such as Civil Aviation Requirements, directives, manuals and advisory circulars and ensure safety oversight of airlines, maintenance organizations, training organizations and other aviation service providers.
- To certify and oversee aerodromes and heliports, including their facilities, operations and safety management systems.

ii. Air navigation and aerodrome services:

- To plan, establish, operate and maintain aerodromes, runways, taxiways, aprons and associated airport infrastructure in Nepal.
- To provide and operate air navigation services, including communication, navigation and surveillance facilities, and air traffic services (other than meteorology, which is normally provided by the Department of Hydrology and Meteorology).
- To install, operate and maintain visual and non-visual aids, air traffic control towers, and related equipment needed for safe and efficient air traffic.

iii. Economic and developmental functions:

- To develop and expand the civil aviation and airport infrastructure of Nepal, including construction, improvement and upgrading of airports and related facilities.
- To prepare plans, programmes and projects for aviation sector development, and to mobilize financial resources, including fees, charges, loans, grants and investment, as allowed by the Act.
- To fix, levy and collect aeronautical and non-aeronautical charges (landing, parking, navigation fees, passenger service charges, rents, concessions) and to manage its own income and expenditure.

iv. Policy advice and coordination:

- To advise the Government of Nepal on civil aviation policy, legislation, regulations and international aviation matters.
- To represent Nepal's civil aviation interests in coordination with government at ICAO and other international aviation forums, subject to government direction.
- To coordinate with other national agencies (tourism, security, immigration, customs, meteorology, etc.) for efficient and safe functioning of airports.

v. Administrative and enforcement powers:

- To frame internal rules, procedures and organizational structures necessary to carry out its functions as authorized by the Act.
- To inspect, investigate and monitor aircraft, operators, aerodromes and aviation personnel, and to enforce compliance by issuing directives, imposing penalties, or suspending operations in case of violations of the Act, regulations or safety requirements.
- To perform any other functions and exercise any powers that are incidental or necessary for fulfilling its statutory mandate as the civil aviation regulator and service provider of Nepal.

7. Briefly point out the special characteristics of airport drainage system. [5]

Answer:

An **airport drainage system** is a system that is designed to collect, convey, and safely dispose of surface runoff and subsurface water from runways, taxiways, aprons, and terminal areas. Surface runoff on the airport pavement can reduce its strength, increase the risk of hydroplaning, and affect aircraft safety. Hence, airport drainage systems require special design considerations beyond those of ordinary road drainage.

Special characteristics of airport drainage system:

a. Rapid removal of surface water:

The drainage system should remove rainwater quickly from runways, taxiways, and aprons. It prevents water accumulation that can cause hydroplaning and reduce braking efficiency.

b. Adequate surface slope:

Pavement surfaces are provided with longitudinal and transverse slopes to facilitate rapid runoff. It ensures no ponding occurs on aircraft movement areas.

c. Subsurface drainage:

Subsurface drains remove groundwater and seepage from beneath pavements.
It prevents weakening of the pavement foundation and improves pavement life.

d. High-capacity drainage network:

Drains, culverts, and channels are designed to carry runoff from intense rainfall without flooding.
The capacity of airport drainage system is based on the selected design storm.

e. Protection of pavement structure:

It prevents water infiltration into pavement layers.
It maintains the strength and durability of runway and taxiway pavements.

f. Erosion control:

Proper lining of drains and channels prevents soil erosion.
It protects embankments and adjacent airport facilities.

g. Flood protection:

The drainage system should protect airport facilities from flooding during heavy rainfall.
Proper outfall arrangements ensure safe disposal of stormwater.

8. List the sequence of layers and test requirements in the field to construct a flexible pavement of parallel taxiway. [2+3=5]

Answer:

A **parallel taxiway** is a taxiway constructed parallel to the runway to facilitate the safe and efficient movement of aircraft between the runway and apron. The flexible pavement of a taxiway consists of several layers placed in sequence from the foundation to the surface. They are described below:

Sequence of layers in flexible pavements:

a. Prepared subgrade:

- Subgrade is the natural soil compacted to the required density.
- It provides the foundation for the pavement.

b. Sub-base course:

- It is the granular material placed over the subgrade.
- Sub-base course improves drainage and distributes loads.

c. Base course:

- Base course consists of high-quality crushed aggregate or Wet Mix Macadam (WMM).
- It acts as the principal structural layer.

d. Prime coat:

- Prime coat is applied on the granular base before laying the bituminous layer.
- It promotes bonding and seals the base surface.

e. Binder course:

- Binder course consists of bituminous mix providing structural strength.
- It distributes aircraft wheel loads.

f. Tack coat:

- Tack coat is applied between bituminous layers.
- It ensures proper bonding between successive asphalt layers.

g. Surface coat:

- Surface coat is the final riding surface for aircraft.
- It provides smoothness, skid resistance, durability, and waterproofing.

Different test requirements in the field to construct the flexible pavement:

During construction, various field tests are carried out to ensure that each layer meets the required quality, strength, and compaction standards. Quality control tests are carried out at every stage of construction which are listed below:

In subgrade course:

- Field Density Test (Sand Cone/Core Cutter)
- Moisture Content Test
- California Bearing Ratio (CBR) Test

In sub-base course:

- Gradation (Sieve Analysis)
- Moisture Content Test
- Field Density Test

In base course:

- Aggregate Gradation Test
- Los Angeles Abrasion Test
- Aggregate Crushing Value (ACV)
- Flakiness and Elongation Index

In surface course:

- Marshall Stability Test
- Core Density Test
- Thickness Measurement
- Surface Evenness Test (Straight Edge)
- Surface Texture and Skid Resistance Test
- Cross Slope and Level Check

9. What are the stability requirements for the design of retaining walls? Explain with neat sketch. [10]

Answer:

A retaining wall is a structure constructed to retain soil at different ground levels. It resists the lateral earth pressure exerted by the retained soil and provides stability to embankments, highways, railways, and foundations. A retaining wall must be designed to satisfy several stability requirements to ensure safety under all loading conditions.

Stability requirements for retaining walls:

i. Stability against overturning:

The wall should not overturn due to the lateral earth pressure acting on its back.

The resisting moment produced by the weight of the wall and the soil over the heel must be greater than the overturning moment.

Requirement:

Factor of Safety (FOS) against overturning ≥ 2.0

ii. Stability against sliding:

The wall should not slide horizontally due to the lateral earth pressure.

Sliding resistance is provided by friction between the base and the foundation soil and, where necessary, by a shear key.

Requirement:

Factor of Safety (FOS) against sliding ≥ 1.5

iii. Bearing capacity stability:

The maximum pressure beneath the base should not exceed the safe bearing capacity of the foundation soil.

The base pressure should be distributed uniformly to avoid excessive settlement.

Requirement:

Maximum bearing pressure $\leq$ Safe bearing capacity of soil.

iv. Stability against excessive settlements:

The wall should undergo only small and uniform settlement.

Differential settlement may lead to cracking, tilting, or structural failure.

v. Stability against structural failure:

The wall stem, heel, toe, and base slab must safely resist bending moments, shear forces, and compressive stresses.

Proper reinforcement should be provided in reinforced concrete retaining walls.

vi. Stability against uplift and water pressure:

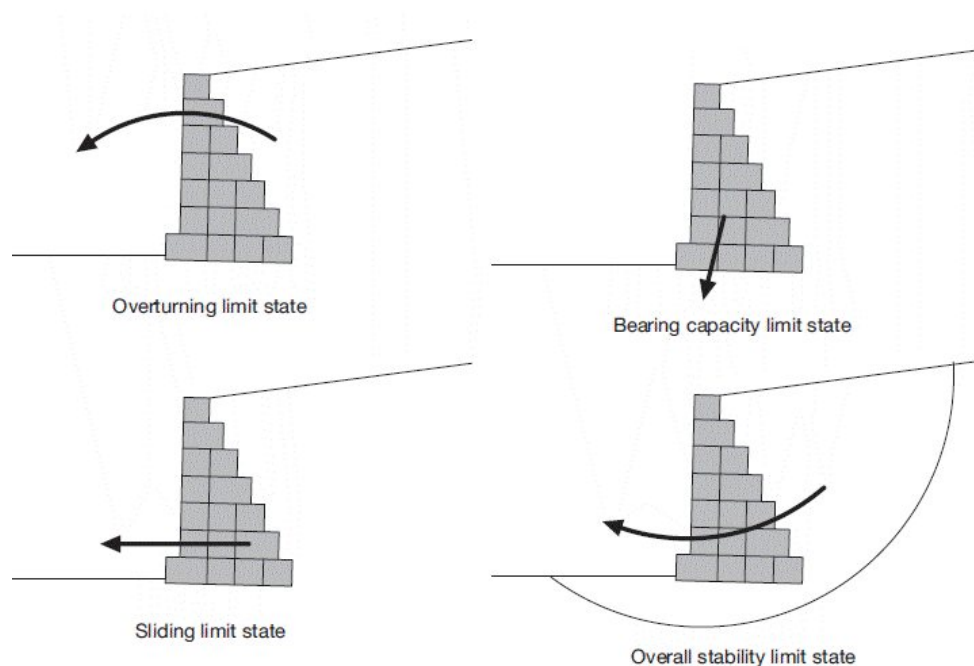
Proper drainage should be provided behind the wall to prevent the buildup of hydrostatic pressure.

Weep holes, drainage blankets, or filter media are commonly used.

vii. Overall stability:

The complete wall–foundation system should remain stable against deep-seated failure or slope failure.

This is especially important for retaining walls constructed on sloping ground or weak foundations.



Hence, the design of a retaining wall must ensure stability against **overturning, sliding, bearing capacity failure, settlement, structural failure, hydrostatic pressure, and overall foundation failure**. Adequate

drainage, proper foundation design, and suitable factors of safety are very important for the long-term performance and safety of the retaining wall.

10. What is contract? Explain general and specific conditions of contract while awarding contract for air traffic control building. [2+8=10]

Answer:

According to Public Procurement Act-2063, a procurement contract is a formal agreement entered into between a public entity and a supplier, construction entrepreneur, consultant, or service provider for the purpose of procurement.

According to International Federation of Consulting Engineers (FIDIC), the Contract is defined as,

"Contract means the Contract Agreement, the Letter of Acceptance, the Letter of Tender, these Conditions, the Specification, the Drawings, the Schedules, and the further documents which are listed in the Contract Agreement or in the Letter of Acceptance."

This definition is taken from the FIDIC Conditions of Contract (1999 Edition, Clause 1.1 – General Provisions).

Some of the elements of valid contract are:

- a. Offer and acceptance
- b. Intention to create legal relationship
- c. Lawful considerations
- d. Competent parties
- e. Free consent
- f. Legal formalities and so on.

Based on the Nepal Public Procurement Act and Regulation, awarding a contract for an air traffic control (ATC) building is categorized as Construction Work, involves both General and Specific Conditions of Contract to manage the project's technical complexity and financial risks.

General Conditions of Contract (GCC)

The General Conditions of Contract (GCC) form a mandatory, standardized section within a Standard Bidding Document (SBD). They establish the foundational legal, financial, and operational rights and obligations of both the contracting parties. The clauses in the GCC are generic and must not be modified.

- **Core Identifying Information:**

The names/addresses of the parties and their contact persons, the budget code, and the priority order of documents in the contract.

- **Operational Terms:**

The scope of work, the work performance schedule, and provisions for site handover to the contractor.

- **Financial Terms:**

The total contract amount or procedure for determining it, and conditions for payment (including mode and currency).

- **Risk and Liability:**

Provisions for Force Majeure (unforeseeable events), insurance requirements, and sub-contracting permissions.

- **Performance Management:**

Terms for Liquidated Damages if the contractor fails to meet deadlines, as well as potential bonuses for early completion.

- **Contract Changes:**

Provisions for issuing Variation Orders (scope changes) and Price Adjustments for inflation or material cost fluctuations.

- **Legal Resolution:**

A mechanism for dispute settlement (such as mutual consent or arbitration) and the applicable law governing the contract.

- **Termination:**

Specific grounds under which the entity or contractor may terminate the agreement, for convenience or breach of contract.

Specific (Special) Conditions of Contract (SCC)

SCC (Special Conditions of Contract) is a critical section that customizes the standard legal framework for a specific project. The Regulation allows a public entity to specify additional conditions in the bidding documents to suit the unique requirements of the project, provided they do not substantially deviate from the GCC. It is also known as Particular Conditions of Contract (PCC). For an ATC building, these would likely include:

- **Technical Specifications:**

Detailed descriptions of the structure's objective, physical nature of the site, and special requirements for mechanical, sanitary, plumbing, and electrical installations critical for ATC operations.

- **Equipment and Maintenance:**

Detailed descriptions of specialized equipment to be supplied, expectations for performance or quality guarantees, and requirements for equipment operating manuals.

- **Insurance Coverage:**

Mandatory insurance for the full replacement cost of the work, materials, and machinery, as well as liability for third-party injury or death.

- **Advanced Safety and Testing:**

Specific stages for inspection and testing, including a completion test and an appropriateness examination of the structure before it is accepted as operational.

- **Site-Specific Notices:**

A requirement for the contractor to post a notice board at the construction site detailing the project name, cost, time schedule, and contact info.

- **Post-Completion Responsibilities:**

Submitting as-built drawings within 30 days of completion and managing a defects liability period before the final return of retention money.

Awarding Mechanism:

Contracts for such structures are generally awarded via open bidding to the lowest evaluated substantively responsive bid. The entity must verify the contractor's technical and financial capacity, including equipment

availability and past performance, to ensure they can handle the complexity of an ATC facility. Upon selection, the contractor must furnish a performance security (typically from a commercial bank) before signing the final procurement contract.

11. What are the different construction equipment and plants used in airport construction? Describe about the different factors affecting the selection of construction equipment. [3+7=10]

Answer:

Airport construction requires specialized equipment and plants for earthwork, pavement construction, concrete works, asphalt production, drainage, and material handling etc. The major construction equipment and plants are as follows:

i. Earthwork equipment:

Used for site clearing, excavation, cutting, filling, and grading.

- **Bulldozer:** Clearing vegetation, pushing soil, rough grading.
- **Excavator:** Excavation of soil, trenches, and foundations.
- **Backhoe Loader:** Small-scale excavation and loading.
- **Motor Grader:** Fine grading and preparation of runway subgrade.
- **Scraper:** Cutting, hauling, and spreading earth over long distances.
- **Dump Truck (Tipper):** Transportation of earth, aggregates, and construction materials.



ii. Compaction equipment:

Used to achieve the required density of soil and pavement layers.

- **Smooth Wheel Roller:** Compaction of asphalt and granular materials.
- **Vibratory Roller:** Compaction of soil and aggregates.
- **Pneumatic Tyre Roller:** Final compaction of asphalt layers.
- **Sheep Foot Roller:** Compaction of cohesive soils.

Types of Rollers



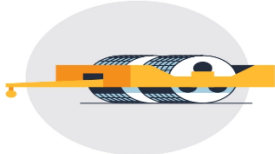
Self-Propelled Roller



Smooth Roller



Pneumatic Tire Roller



Grid Roller



Sheepfoot Roller



Track Roller

iii. Asphalt pavement equipment:

Used in runway, taxiway, and apron construction.

- **Asphalt Mixing Plant (Hot Mix Plant):** Produces asphalt concrete.
- **Bitumen Sprayer (Distributor):** Applies prime coat and tack coat.
- **Asphalt Paver Finisher:** Lays asphalt pavement uniformly.
- **Chip Spreader:** Spreads aggregates for surface treatment.

iv. Crushing and screening plants:

Used for aggregate production.

- **Stone Crusher Plant** – Crushes rocks into aggregates.
- **Screening Plant** – Separates aggregates by size.
- **Aggregate Washing Plant** – Cleans aggregates before use.

The selection of construction equipment is an important decision that affects the cost, quality, productivity, and timely completion of a project. The following factors should be considered while selecting construction equipment:

i. Nature and type of project:

Type of work such as earthwork, pavement, concrete, airport, bridge, etc. affects the selection of the construction equipment.

Project size and complexity also determine the type and capacity of equipment required

ii. Site conditions:

Topography (plain, hilly, rocky, or marshy terrain).

Soil conditions and bearing capacity.

Site accessibility and available working space.

Climatic conditions such as rainfall, temperature, and wind.

iii. Equipment capacity and productivity:

Equipment should have adequate capacity to meet the required production rate.

Proper matching of equipment capacities improves efficiency and reduces idle time.

iv. Project duration and schedule:

Short-duration projects may favor hiring equipment.
Long-duration projects may justify purchasing equipment.
Equipment should help meet project deadlines.

v. Cost of equipment:

Initial purchase price.
Operating costs (fuel, lubrication, and operator wages).
Maintenance and repair costs.
Depreciation, insurance, and resale value.

vi. Availability of equipment:

Availability in the local market.
Ease of obtaining spare parts and technical support.
Availability of rental or leasing options.

vii. Availability of skilled operators:

Equipment should match the availability of trained operators and maintenance personnel.
Highly sophisticated equipment requires specialized training.

viii. Environmental considerations:

Fuel efficiency and emissions.
Noise and dust generation.
Compliance with environmental regulations.

12. Discuss the components and functions of Pavement Management System (PMS) in maintaining airport pavements. Write techniques for evaluating the structural strength of pavements. How do you assess runway surface conditions? Mention. [4+4+2=10]

Answer:

A pavement management system is a structured way to monitor, evaluate, and plan maintenance and rehabilitation for roads, streets, runways, and other paved assets. It helps owners decide where to spend limited funds to keep pavements in good condition for as long as possible.

Components of a pavement Management System (PMS)

i. Pavement inventory:

Collection of basic information about airport pavements.
Includes:

- a. Pavement location and identification
- b. Dimensions and geometry
- c. Pavement type (flexible or rigid)
- d. Construction history
- e. Material properties
- f. Traffic information

ii. Pavement condition assessment:

Regular inspection of pavement surface.
Includes:

- a. Type of distress (cracks, rutting, potholes, spalling, FOD risk)
- b. Severity of distress
- c. Extent of damage

Uses indices such as the Pavement Condition Index (PCI).

iii. Structural evaluation:

Determines the structural capacity of the pavement.

Methods include:

- a. Deflection testing (e.g., Falling Weight Deflectometer)
- b. Core sampling
- c. Load-bearing capacity evaluation

iv. Traffic and loading system:

Includes:

- a. Aircraft types
- b. Aircraft weight
- c. Number of movements
- d. Wheel load configurations

Essential for predicting pavement deterioration.

Functions of pavement management system:

- i. It helps to collect the data of pavement conditions through inspection.
- ii. It uses the data to rate the pavement health by using indexes such as PCI (Pavement condition Index)
PCI = 85–100: Excellent
PCI = 70–85: Good
PCI = 55–70: Fair
PCI = Below 55: Poor to Failed
- iii. It predicts the deterioration rate and need for repair and maintenance of pavements.
- iv. The system helps to prioritize repairs, preservation, resurfacing and reconstruction.

The structural strength of airport pavements is evaluated to determine whether the pavement can safely withstand aircraft loads without excessive deformation or failure. The following techniques are commonly used:

i. Non-Destructive Testing (NDT) Methods:

a. Falling weight Deflectometer:

Applies a dynamic load simulating an aircraft wheel load.
Measures pavement deflection using sensors.
Estimates pavement layer stiffness and remaining structural capacity.
Most widely used method for airport pavements.

b. Benkelman Beam Test:

Measures pavement rebound deflection under a standard wheel load.
Commonly used for flexible pavements.
Indicates structural adequacy and overlay requirements.

c. Group Penetrating Radar (GPR):

Uses electromagnetic waves to determine pavement layer thickness.
Detects voids, moisture, and subsurface defects without excavation.

ii. Destructive Testing Methods:

a. Core sampling:

Cylindrical cores are extracted from pavement.
Determines pavement thickness, density, material quality, and condition.

b. Test pits:

Excavation to inspect pavement layers.

Used to determine actual layer thickness and material condition.

c. Laboratory testing:

Tests on collected samples include:

- a. Compressive strength
- b. Marshall Stability (bituminous mixes)
- c. CBR
- d. Resilient modulus
- e. Density

iii. Analytical Evaluation:

Evaluation of **Aircraft Classification Number (ACN)** and **Pavement Classification Number (PCN)**.

Comparison of aircraft loading with pavement structural capacity.

Life-cycle and fatigue analysis.

Assessment of Runway Surface Conditions:

Runway surface conditions are assessed regularly to ensure safe aircraft take-off and landing. The following methods are used:

i. Visual inspection:

Inspection for cracks, rutting, raveling, potholes, joint failure, Foreign Object Debris (FOD), surface deformation.

ii. Pavement Condition Index (PCI):

Standard method for evaluating pavement condition.

Rates pavement from **0 to 100**:

- a. **85–100**: Excellent
- b. **70–84**: Good
- c. **55–69**: Fair
- d. **40–54**: Poor
- e. **Below 40**: Very Poor/Failed

iii. Drainage measurement:

Checks cross slope, longitudinal slope, and drainage systems.

Identifies areas prone to water ponding or poor runoff.

iv. Surface texture measurement:

Evaluates macrotexture and microtexture.

Ensures adequate drainage and skid resistance.